Type MS · Link a nutrient to its function, source and deficiency disease

Answer 1 of 13 · Question 2 part b · 2 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| **2(a)** | Any two from:<br><br>carbohydrates/proteins/lipids/mineral salts (or named example)/fibre;; | Ignore named examples<br><br>(1 mark for each two correct) | **(2)** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **2(b)** | Any two from:<br>• lack of it causes scurvy;<br>• (lack of vitamin C causes) damage to membranes/bleeding gums;<br>• makes connective tissue/maintains/strengthens cell linings/gums/sticks/joins cells/tissues together; | | **(2)** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **2(c)(i)** | Any five from:<br><br>• Add fresh juice/long-life juice (to a test tube);<br>• add DCPIP drop by drop/slowly;<br>• until blue colour disappears/solution turns colourless;<br>• count number of drops/measure volume of DCPIP added;<br>• repeat (to obtain mean);<br>• compare volumes of DCPIP added; | | **(5)** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **2(c)(ii)** | Any two from:<br>• volume of juice;<br>• temperature;<br>• concentration DCPIP; | Allow amount for mp1. Ignore concentration | **(2)** |

**Total 11 marks**

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 1 of 13 · Question 2 part b · 2 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

**Paper 1: Subject Human Biology**

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 2 of 13 · Question 4 part c · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

| | | | |
|---|---|---|---|
| 4 (a) | $\frac{30\,000}{100}$ x 40;<br><br>12 000; | Full marks for final correct answer<br><br>ecf | 1<br><br>1 |
| (b) | a microorganism/named microorganism that causes disease; | | 1 |
| (c) | It cannot be transmitted from one person to the next | | 1 |
| (d) | • use a sample of people/use two groups of people;<br>• feed each sample/group of people a different diet/feed one group a diet with vitamin B and the other without vitamin B;<br>• monitor health/compare (health) of two groups; | | 1<br><br>1<br><br>1 |
| (e) | • breakdown of cartilage/cartilage not formed properly/damaged;<br>• bones rub together/more friction between bones/less shock absorption; | | 1<br><br>1 |

**Total question 4 = 9 marks**

# Type MS · Link a nutrient to its function, source and deficiency disease

## Answer 2 of 13 · Question 4 part c · 1 mark

2019 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 3 of 13 · Question 1 part b.i · 1 mark

2020 JAN · Paper 01 · find part b.i in this mark scheme extract

| number | | | | |
|---|---|---|---|---|
| 1 | (a) | **The only correct answer is C** to slow down the growth of bacteria<br><br>*A is incorrect because taste is personal preference*<br><br>*B is incorrect because it does not kill the bacteria*<br><br>*D is incorrect because the appearance is not affected* | | 1 |
| | (b) (i) | protein | | 1 |
| | (ii) | **The only correct answer is D** it does not contain energy from fat<br><br>*A is incorrect because vitamins have no effect on body mass*<br><br>*B is incorrect because cholesterol has no effect on body mass*<br><br>*C is incorrect because it does contain energy from carbohydrates* | | 1 |
| | (c) | • add Benedict's solution (to a sample of yogurt) (1)<br>• heat gently (1)<br>• brick red colour indicates the presence of sugar (1) | R if adding to glucose/reducing sugar<br><br>Allow yellow, green, orange | 3 |

**Total for Question 1 = 6 marks**

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 4 of 13 · Question 4 part c.i · 1 mark

2020 JAN · Paper 02 · find part c.i in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 4 | (a)(i) | pinworms/eggs in faeces/nutritional deficiency/named deficiency/weight loss/malnutrition(1) | Accept fatigue | 1 |
| | (ii) | Any two from:<br>• itching transfers eggs (from anus) to fingers/hand(1)<br>• eggs transferred from fingers to other surfaces/people(1)<br>• eggs ingested when other people touch (contaminated) surfaces (and put hands in their mouth)(1) | | 2 |
| | (b) | Any two from:<br>• correct temperature/pH/provides warmth(1)<br>• plenty of nutrients(1)<br>• no predators(1) | Allow food | 2 |
| | (c) (i) | lacking in one or more nutrients/vitamins/minerals / other named nutrient(1) | | 1 |
| | (ii) | any named nutritional deficiency such as anaemia/scurvy | Allow named nutrient e.g. protein deficiency | 1 |
| | (iii) | • pinworms absorb nutrients/named nutrient(1)<br>• less nutrients absorbed/pass into blood/body/cells receive less nutrients(1) | Allow food | 2 |
| | (d) | Reference to eggs/pinworms on transparent tape (where they can be seen) | | 1 |
| | (e) | Any two from:<br>• refrain from itching/scratching (anus)(1)<br>• use medication (to kill pinworms)(1)<br>• don't share towels/clothing(1)<br>• wash bedclothes/linen regularly/do not shake bed linen(1)<br>• don't bite nails/keep fingernails short / clean/wash hands (after using toilet)(1)<br>• don't allow children to bathe together(1) | | 2 |

**Total for Question 4 = 12 marks**

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 5 of 13 · Question 4 part c.ii · 1 mark

2020 JAN · Paper 02 · find part c.ii in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 4 | (a)(i) | pinworms/eggs in faeces/nutritional deficiency/named deficiency/weight loss/malnutrition(1) | Accept fatigue | 1 |
| | (ii) | Any two from:<br>• itching transfers eggs (from anus) to fingers/hand(1)<br>• eggs transferred from fingers to other surfaces/people(1)<br>• eggs ingested when other people touch (contaminated) surfaces (and put hands in their mouth)(1) | | 2 |
| | (b) | Any two from:<br>• correct temperature/pH/provides warmth(1)<br>• plenty of nutrients(1)<br>• no predators(1) | Allow food | 2 |
| | (c) (i) | lacking in one or more nutrients/vitamins/minerals / other named nutrient(1) | | 1 |
| | (ii) | any named nutritional deficiency such as anaemia/scurvy | Allow named nutrient e.g. protein deficiency | 1 |
| | (iii) | • pinworms absorb nutrients/named nutrient(1)<br>• less nutrients absorbed/pass into blood/body/cells receive less nutrients(1) | Allow food | 2 |
| | (d) | Reference to eggs/pinworms on transparent tape (where they can be seen) | | 1 |
| | (e) | Any two from:<br>• refrain from itching/scratching (anus)(1)<br>• use medication (to kill pinworms)(1)<br>• don't share towels/clothing(1)<br>• wash bedclothes/linen regularly/do not shake bed linen(1)<br>• don't bite nails/keep fingernails short / clean/wash hands (after using toilet)(1)<br>• don't allow children to bathe together(1) | | 2 |

**Total for Question 4 = 12 marks**

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 6 of 13 · Question 4 part a · 2 marks

2020 NOV · Paper 01 (regional R) · find part a in this mark scheme extract

| | | | |
|---|---|---|---|
| 4 (a) | • ankles(1)<br>• protein(1) | | 2 |
| (b) (i) | • axes labels(1)<br>• axes scales(1)<br>• correctly plotted bars(1) | Max 2 marks for line graph | 3 |
| (ii) | improved/varied/balanced diet/education (in nutrition) / international aid/less protein in diet in 2010(1) | | 1 |
| (iii) | • 95 ÷ 1978 (x 100) (1)<br>• 4.8(%) | Full marks for correct final answer | 2 |

**Total for Question 4 = 8 marks**

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 7 of 13 · Question 4 part b.ii · 1 mark

2020 NOV · Paper 01 (regional R) · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 4 (a) | • ankles(1)<br>• protein(1) | | 2 |
| (b) (i) | • axes labels(1)<br>• axes scales(1)<br>• correctly plotted bars(1) | Max 2 marks for line graph | 3 |
| (ii) | improved/varied/balanced diet/education (in nutrition) / international aid/less protein in diet in 2010(1) | | 1 |
| (iii) | • 95 ÷ 1978 (x 100) (1)<br>• 4.8(%) | Full marks for correct final answer | 2 |

**Total for Question 4 = 8 marks**

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 8 of 13 · Question 5 part a.i · 1 mark

2020 NOV · Paper 01 (regional R) · find part a.i in this mark scheme extract

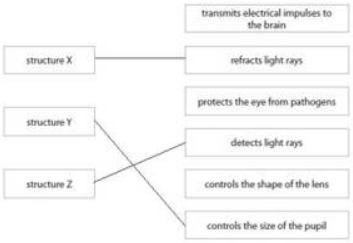
| | | | | |
|---|---|---|---|---|
| 5 (a) | (i) | **C** green vegetables and red meat | | 1 |
| | (ii) | • (iron) needed for production of haemoglobin(1)<br>• (haemoglobin/red blood cell) binds to/transports oxygen(1)<br>• oxygen needed for (aerobic) respiration/energy(1)<br>• for growth of fetus(1) | | 4 |
| (b) | (i) | Any four from:<br>• take temperature of water (at the start) (1)<br>• reference to measuring mass of food(1)<br>• place food on holder(1)<br>• burn food(1)<br>• reference to time (for burning food)(1)<br>• take temperature of water after burning/once food is completely burnt/measure change in temperature(1) | Allow amount for mass<br>Allow food is heated/set alight | 4 |
| | (ii) | Any three from:<br>• add a lid(1)<br>• reference to stirring/even heat distribution(1)<br>• insulate the copper can/move food closer (to the can/water) (1)<br>• reference to preventing heat loss(1) | Allow cover the beaker | 3 |

**Total for question 5 = 12 marks**

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 9 of 13 · Question 1 part b.iv · 1 mark

2023 JAN · Paper 01 · extract 1 of 2 · find part b.iv in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 1 | (a) | (i) |  | | 3 |
| | | (ii) | A;<br>B, C and D do not describe stereoscopic vision. | | 1 |
| | (b) | (i) | build-up of protein/cloudy/opaque;<br>in the lens; | | 2 |
| | | (ii) | surgery/operation to replace the lens/lens replacement; | | 1 |
| | | (iii) | 39 000000 x 43 ÷ 100;<br>16.77 million/16.8 million/16 770 000; | | 2 |
| | | (iv) | vitamin A; | | 1 |

Total 10 marks

## Type MS · Link a nutrient to its function, source and deficiency disease

### Answer 9 of 13 · Question 1 part b.iv · 1 mark

2023 JAN · Paper 01 · extract 2 of 2 · find part b.iv in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Link a nutrient to its function, source and deficiency disease

### Answer 10 of 13 · Question 4 part d · 5 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part d in this mark scheme extract

| Question number | | | | |
|---|---|---|---|---|
| 4 | (a) | • correct plots;;;<br>• key/labels; | | 4 |
| | (b) | • correct components/nutrients/list of components min of five;<br>• in correct amounts/proportions; | | 2 |
| | (c) | • 75;<br>• 15<br>• = 5:1; | ECF | 2 |
| | (d) | • contain protein<br>• needed for growth/repair/,muscles/tissues;<br>• area A girl has low(er) levels/area B girl has high(er) levels;<br>• less/poor growth/area B girl good growth;<br>• kwashiorkor; | | 5 |

Total 13 marks

## Type MS · Link a nutrient to its function, source and deficiency disease

### Answer 10 of 13 · Question 4 part d · 5 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part d in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 11 of 13 · Question 3 part a.iv · 1 mark

2024 JUN · Paper 01 (regional R) · find part a.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (i) | contains all nutrients needed/ correct balance of nutrients needed/ correct proportion of nutrients;<br>to maintain health; | If groups of nutrients given all must be named | (2) |
| (ii) | males need more energy than females;<br>3000 kJ;<br>greater muscle mass;<br>males are likely to be more active/ higher metabolic rate: | Allow reverse for female | (3) |
| (iii) | iron needed to form haemoglobin;<br>to replace haemoglobin lost during menstruation/red blood cells are lost during menstruation; | | (2) |
| (iv) | vitamin A; | | (1) |
| (v) | In 250g red meat - 2.5 x 2.5 = 6.25;<br>In 200g broccoli - 2 x 0.7 = 1.4;<br>12 - 7.65;<br>4.35 (mg); | Full marks for correct final answer<br><br>ecf for mp3 and mp4 | (4) |
| (b) (i) | bleeding gums/anaemia/bruising/poor wound healing; | Allow other valid symptoms | (1) |
| (ii) | more fruit/vegetables eaten in diet/increased intake of vitamin C/ vitamin c supplement; | | (1) |

**Total for Question 3 = 14 marks**

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 12 of 13 · Question 3 part b.i · 1 mark

2024 JUN · Paper 01 (regional R) · find part b.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (i) | contains all nutrients needed/ correct balance of nutrients needed/ correct proportion of nutrients; to maintain health; | If groups of nutrients given all must be named | (2) |
| (ii) | males need more energy than females;<br>3000 kJ;<br>greater muscle mass;<br>males are likely to be more active/ higher metabolic rate: | Allow reverse for female | (3) |
| (iii) | iron needed to form haemoglobin;<br>to replace haemoglobin lost during menstruation/red blood cells are lost during menstruation; | | (2) |
| (iv) | vitamin A; | | (1) |
| (v) | In 250g red meat - 2.5 x 2.5 = 6.25;<br>In 200g broccoli - 2 x 0.7 = 1.4;<br>12 - 7.65;<br>4.35 (mg); | Full marks for correct final answer<br><br>ecf for mp3 and mp4 | (4) |
| (b) (i) | bleeding gums/anaemia/bruising/poor wound healing; | Allow other valid symptoms | (1) |
| (ii) | more fruit/vegetables eaten in diet/increased intake of vitamin C/ vitamin c supplement; | | (1) |

**Total for Question 3 = 14 marks**

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 13 of 13 · Question 3 part b.ii · 1 mark

2024 JUN · Paper 01 (regional R) · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (i) | contains all nutrients needed/ correct balance of nutrients needed/ correct proportion of nutrients; to maintain health; | If groups of nutrients given all must be named | (2) |
| (ii) | males need more energy than females;<br>3000 kJ;<br>greater muscle mass;<br>males are likely to be more active/ higher metabolic rate: | Allow reverse for female | (3) |
| (iii) | iron needed to form haemoglobin;<br>to replace haemoglobin lost during menstruation/red blood cells are lost during menstruation; | | (2) |
| (iv) | vitamin A; | | (1) |
| (v) | In 250g red meat - 2.5 x 2.5 = 6.25;<br>In 200g broccoli - 2 x 0.7 = 1.4;<br>12 - 7.65;<br>4.35 (mg); | Full marks for correct final answer<br><br>ecf for mp3 and mp4 | (4) |
| (b) (i) | bleeding gums/anaemia/bruising/poor wound healing; | Allow other valid symptoms | (1) |
| (ii) | more fruit/vegetables eaten in diet/increased intake of vitamin C/ vitamin c supplement; | | (1) |

**Total for Question 3 = 14 marks**